Домашняя работа 33 TMS по теме Google Cloud S3 Bucket + IAM.

1) Создание сервисного аккаунта с ролью Storage.Admin:

1.1)создаем сервис аккаунт

cloudshell:~ (it-server-344307)$ `gcloud iam service-accounts create sa-tmsdz33 --display-name "TMS service account"`

`Created service account [sa-tmsdz33].`

1.2)добавляем сервис-аккаунту роль Storage.Admin

cloudshell:~ (it-server-344307)$ `gcloud projects add-iam-policy-binding $DEVSHELL_PROJECT_ID --member serviceAccount:sa-tmsdz33@$DEVSHELL_PROJECT_ID.iam.gserviceaccount.com --role="roles/storage.admin"`

```
Updated IAM policy for project [it-server-344307].
bindings:
- members:
  - serviceAccount:service-52615052819@containerregistry.iam.gserviceaccount.com
  role: roles/containerregistry.ServiceAgent
- members:
  - serviceAccount:52615052819-compute@developer.gserviceaccount.com
  - serviceAccount:52615052819@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
  - user:egor.belousov@altezza.org
  role: roles/owner
- members:
  - serviceAccount:gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com
  - serviceAccount:sa-tmsdz33@it-server-344307.iam.gserviceaccount.com
  role: roles/storage.admin
version: 1
```

1.3)создаем ключ доступа

cloudshell:~ (it-server-344307)$ **PROJECT_ID="$DEVSHELL_PROJECT_ID"** // (it-server-344307 - ID проекта).
cloudshell:~ (it-server-344307)$ **SA_NAME="sa-tmsdz33"**
cloudshell:~ (it-server-344307)$ **gcloud config set project it-server-344307** // указываем проект
cloudshell:~ (it-server-344307)$ **gcloud iam service-accounts keys create ~/keys/${SA_NAME}.json --iam-account=${SA_NAME}@${PROJECT_ID}.iam.gserviceaccount.com**
created key [eda440aa13d132d9538138b5d4fc28ab74d11678] of type [json] as [/home/egor_v/keys/sa-tmsdz33.json] for [sa-tmsdz33@it-server-344307.iam.gserviceaccount.com]

P.S.#1 или так **gcloud iam service-accounts keys create key.json --iam-account=gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com (без переменных)**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud iam service-accounts keys create key.json \
    --iam-account=gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com
created key [1c1e5cad0bfe780800b579c73dc60971f0810611] of type [json] as [key.json] for [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
```

1.3.2) // создаем 2ю сервисную учетку “gitlab-runner-sa” (для теста) :

cloudshell:~ (it-server-344307)$ **gcloud iam service-accounts create gitlab-runner-sa   --description="Service account for GitLab runner on U22GITLABNODE2"   --display-name="GitLab Runner SA"**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud iam service-accounts create gitlab-runner-sa \
    --description="Service account for GitLab runner on U22GITLABNODE2" \
    --display-name="GitLab Runner SA"
Created service account [gitlab-runner-sa].
```

1.3.3) Проверяем созданные IAM-учетки в нашем проекте :

cloudshell:~ (it-server-344307)$ **gcloud iam service-accounts list**
DISPLAY NAME: Compute Engine default service account
EMAIL: 52615052819-compute@developer.gserviceaccount.com
DISABLED: False

DISPLAY NAME: GitLab Runner SA
EMAIL: gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com
DISABLED: False

DISPLAY NAME: TMS service account
EMAIL: sa-tmsdz33@it-server-344307.iam.gserviceaccount.com
DISABLED: False

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud iam service-accounts list
DISPLAY NAME: Compute Engine default service account
EMAIL: 52615052819-compute@developer.gserviceaccount.com
DISABLED: False

DISPLAY NAME: GitLab Runner SA
EMAIL: gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com
DISABLED: False

DISPLAY NAME: TMS service account
EMAIL: sa-tmsdz33@it-server-344307.iam.gserviceaccount.com
DISABLED: False
egor_belousov@cloudshell:~ (it-server-344307)$
```

1.4) копируем ключ (json-файл) на удаленный хост:

@cloudshell:~/keys (it-server-344307)$ **cat sa-tmsdz33.json**
{
  "type": "service_account",

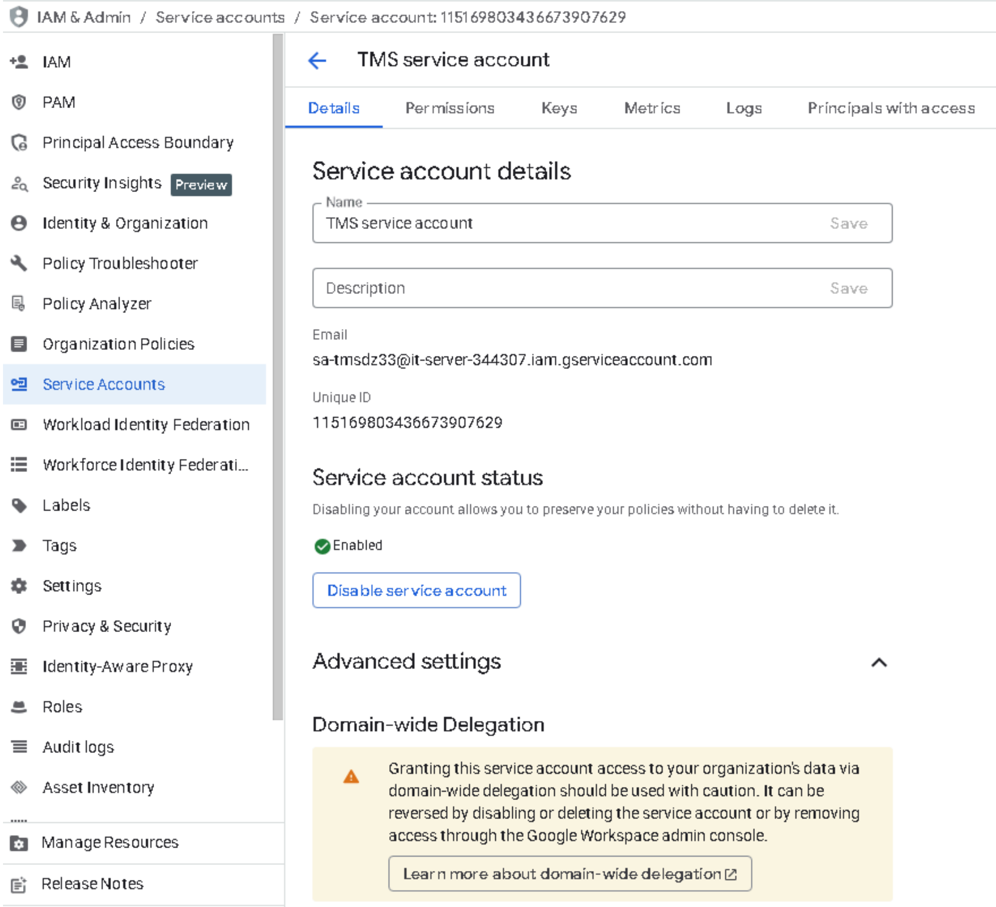
"project_id": "it-server-344307",
"private_key_id": "eaa13d132d9538138b5d4fc28ab74d11678",
"private_key": "-----BEGIN PRIVATE KEY-----\nMIIEvwIBADANBgkqhkiG9w0BAQEFAASCBKkwggSlAgEAAoIBAQDAPRiJ+/yV79B1\nbZOTJz5oXHfpayfKTo6PBOdOibM97OTFKvCmA94dXRDLbL2MIQWuOW4RKe8wK6ZL\ncmUr97/f9spqQ/pksN4KwAUXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXQcuNGbg==\n-----END PRIVATE KEY-----\n",
"client_email": "sa-tmsdz33@it-server-344307.iam.gserviceaccount.com",
"client_id": "115169803436673907629",
"auth_uri": "https://accounts.google.com/o/oauth2/auth",
"token_uri": "https://oauth2.googleapis.com/token",
"auth_provider_x509_cert_url": "https://www.googleapis.com/oauth2/v1/certs",
"client_x509_cert_url": "https://www.googleapis.com/robot/v1/metadata/x509/sa-tmsdz33%40it-server-344307.iam.gserviceaccount.com",
"universe_domain": "googleapis.com"
}

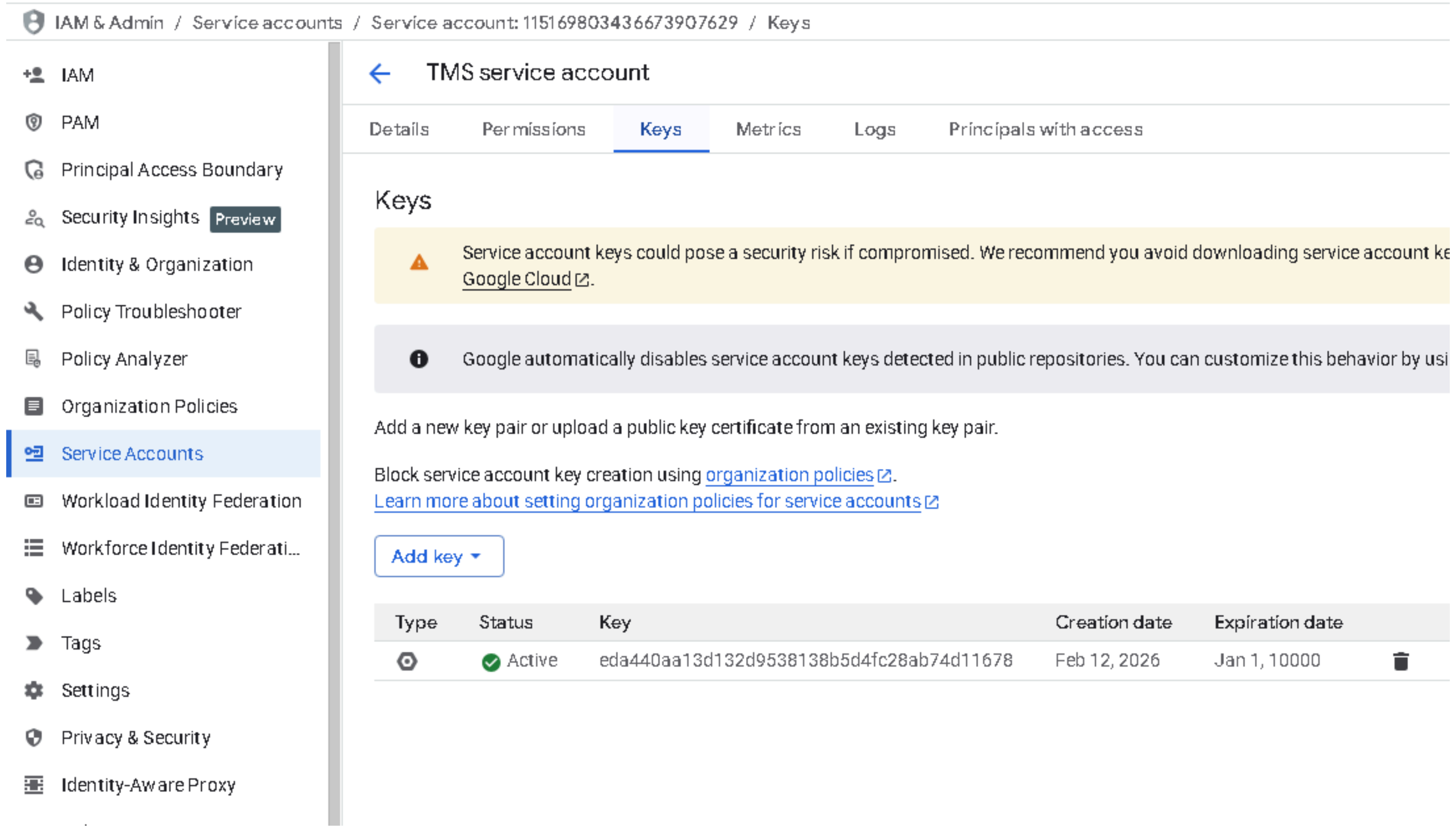
1.5) авторизоваться на ubuntu22 по key.json-файлу :

```
ubuntu22# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com   --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
```

```
root@U22GITLABNODE2:/home/gitlab-runner# nano key.json
root@U22GITLABNODE2:/home/gitlab-runner# cat key.json
{
  "type": "service_account",
  "project_id": "it-server-344307",
  "private_key_id": "1c1e5cad0bfe780800b579c73dc60971f0810611",
  "private_key": "-----BEGIN PRIVATE KEY-----\nMIIEvAIBADANBgkqhkiG9w0BAQEFAASCBKYwggSiAgEAAoIBAQD3Us4L-          9rs3HlpzaQ/V0u80B71zxnSg6V3H          Y5ZoA47VFVmhHawLI4olKNla9jsetxzan
b2pa9YRAXVl3vGydc4DM\nwHCTrhSluTuC9bdyYsiVY9V982CrqqpsrQ6glCzwh1FIgQ4dwLvTgHPe/6iAH5X9\ndJbPLYByPK3PqK0:          DSHSoBXAkRY+DKYslw/1u\nkNsOD          DqR+ifob3Z2L8Y8w6opdkVjjSWBEV\nB3w
jyEpBAgMBAAECggEAEDUBStxeSjn3xcjA34CG7sPSvqL5e+bfM0Dz96VKoHtz\n6+fVm8/t1cRu+n/u9aoWOvfvnY4WrKRggbrZr4S2          9MpZkZyW/eq1/OF6ATYa7kViH3Oa          :oOw\ncyFBLmAPJudKrBkByapfTGsfGMIQ
Fh6pVuVm+7qx2QHq/WfoKZgEt7xc45G7z5Az\np057TxrhoP19Ohmdh+s/kkyuAzbyf7hT+CrzyvzqjALYd8nWmmP5Gs8o4FJkPC7f\          7BSiDQKXNow+0mQKBgQD8Edw7ieA          DPLgUw5k9pFiXYZvStflXu35NClru+PwlI
6TCdCYo4tRQ\nbwNU3ixOp/rRk0aoF7bbfHpKYM+uR4nikzeYd80iYflIpi69k5llSa2+sTEdsbbv\nKmCdr/24busSIjVkjnfV++iOaQKBgQD/Lf/npITOBGsVLbQtA9/PpSnRom8Kxj+w\niS1LrOKEIeCa4eM96J/zFjxcJQK0fk1a0/yef2bTuGQ9Usb/idvzG/CYlcyxJHYB\nSGakZEBI67sL
vaNDEq6JMGkuHDfoq9D0/nWUAfKllqzAkis/lYIKhq+7RPZwQpYZ\nEn2o67cSGQKBgDgtitXCHq4sJtON1f3+SRy+/C7ql7vjRXqSdE+WYrhN94Mjc5MZ\nb+8AG3gQE0/8NNP5rchhL0EfPbnzA7qBY06wzVMiYOYAWsjvsBMupNqaxO5khzh1\n8vgKCy/UgghM8THTCWbnis9rEjU5/DlFSo8aQ
AOwxLcf5KJqlg6cu/GxAoGAa8Qi\npWs/kVyH7QA864tbj09Xj9/XMJI6RS21x2AXl4VJbmo4KxmhCL6jiw0+Cd6epbdu\nl7oljfTw3Pak9oJZYga8xVqQBfiUwJwlnaZ9hkvNZo4oe+CKDcY9elEVE7iT4GlE\nXQow5yasW7ra317v+kArKXe/HA8VgxBR5aWsWBECgYA3ed8VMaowUbck5LCjYh
MZ\nelhaOwaZ8AD1ewn0GGTWYCmFITouXZKpxX6noS4U4FL0mD55m5Se2zFw69oJ9lJT\nLD2xv0NznusUQn6cwjuiPTWi7vSJ7cNEbEIO/RFJNiafF+7sfDVlfb9uQ/McfGde\npnUlf2I037ltLzDA0mqw+w==\n-----END PRIVATE KEY-----\n",
  "client_email": "gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com",
  "client_id": "114450695213956486347",
  "auth_uri": "https://accounts.google.com/o/oauth2/auth",
  "token_uri": "https://oauth2.googleapis.com/token",
  "auth_provider_x509_cert_url": "https://www.googleapis.com/oauth2/v1/certs",
  "client_x509_cert_url": "https://www.googleapis.com/robot/v1/metadata/x509/gitlab-runner-sa%40it-server-344307.iam.gserviceaccount.com",
  "universe_domain": "googleapis.com"
}
root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account egor.belousov@altezza.org   --key-file=key.json
ERROR: (gcloud.auth.activate-service-account) Invalid value for [ACCOUNT]: The given account name does not match the account name in the key file.  This argument can be omitted when using .json keys.
root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com   --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
root@U22GITLABNODE2:/home/gitlab-runner#
```

1.6) проверяем наш сервис-аккаунт в Google Cloud :



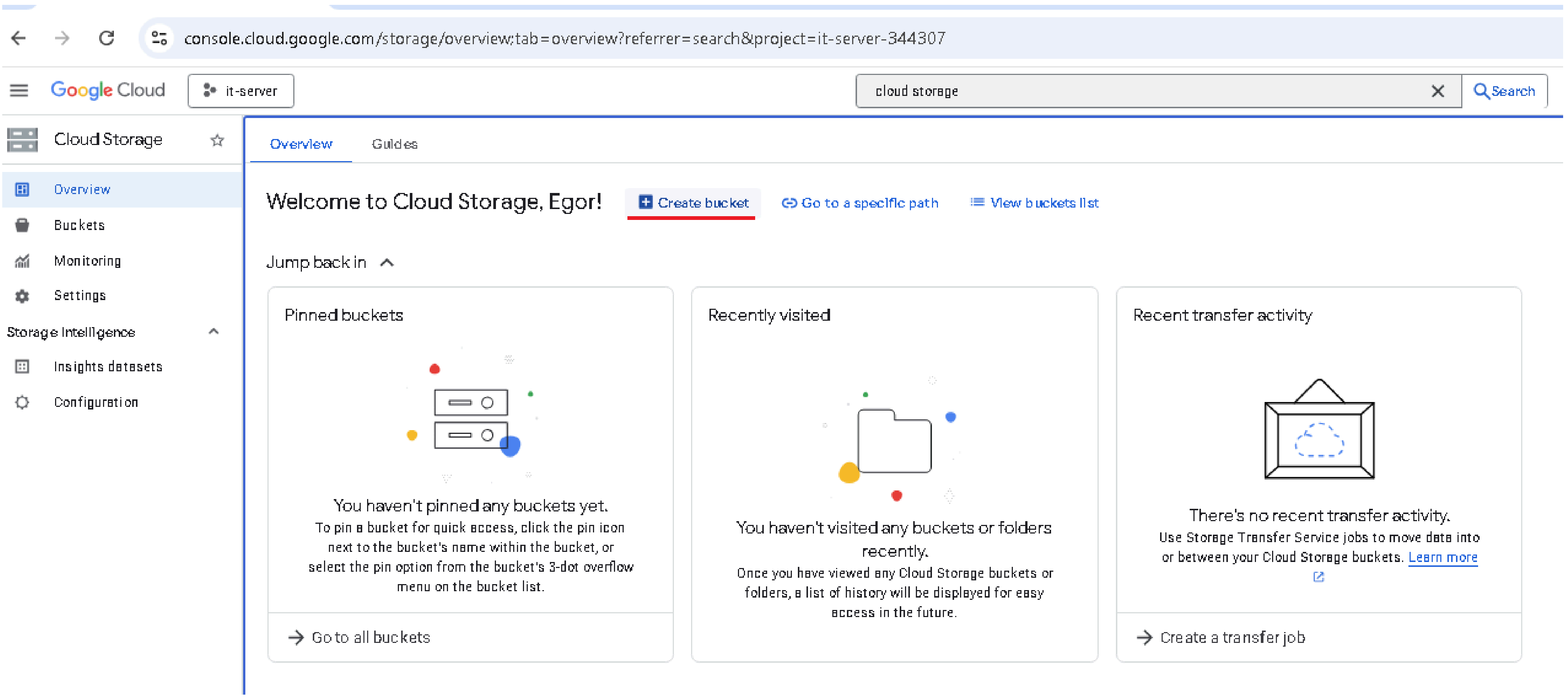


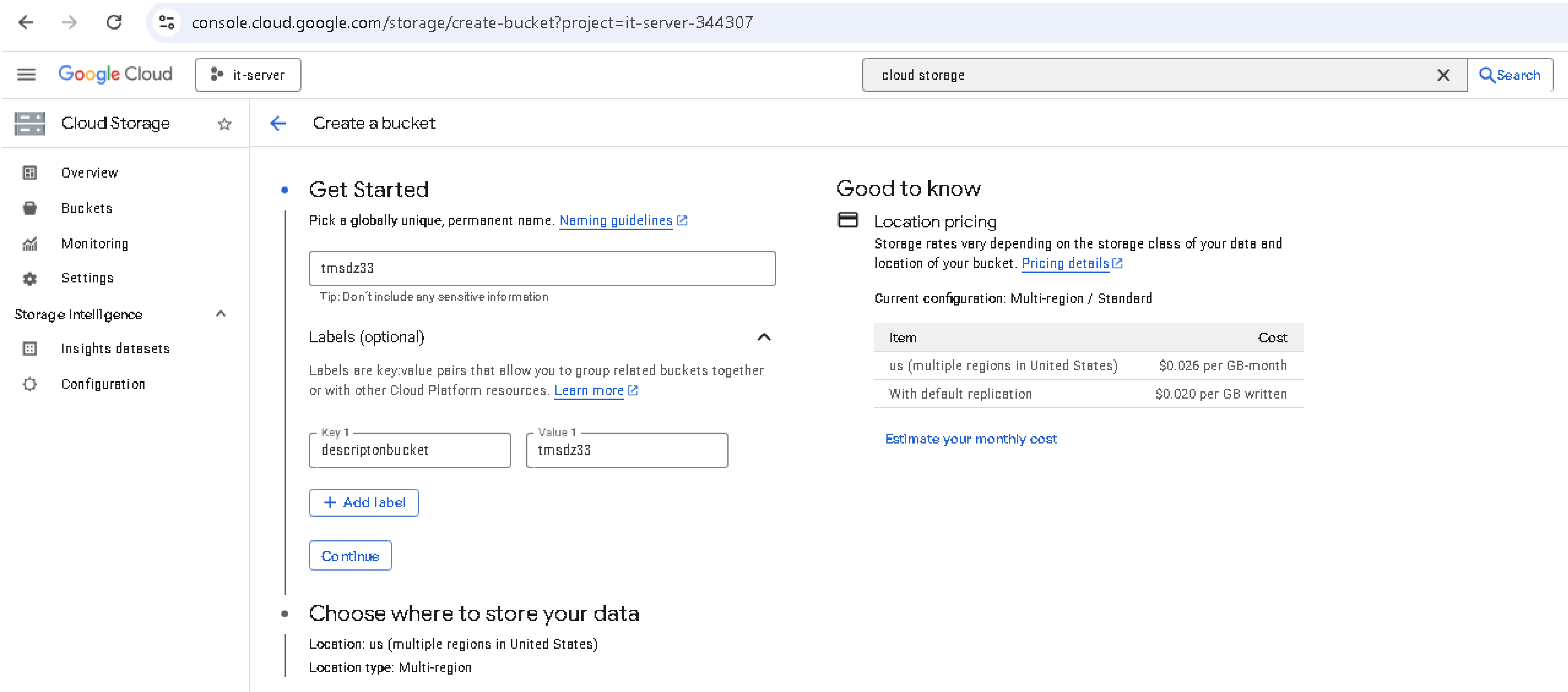
2) Создаем S3 GCP Bucket (cli-вариант)

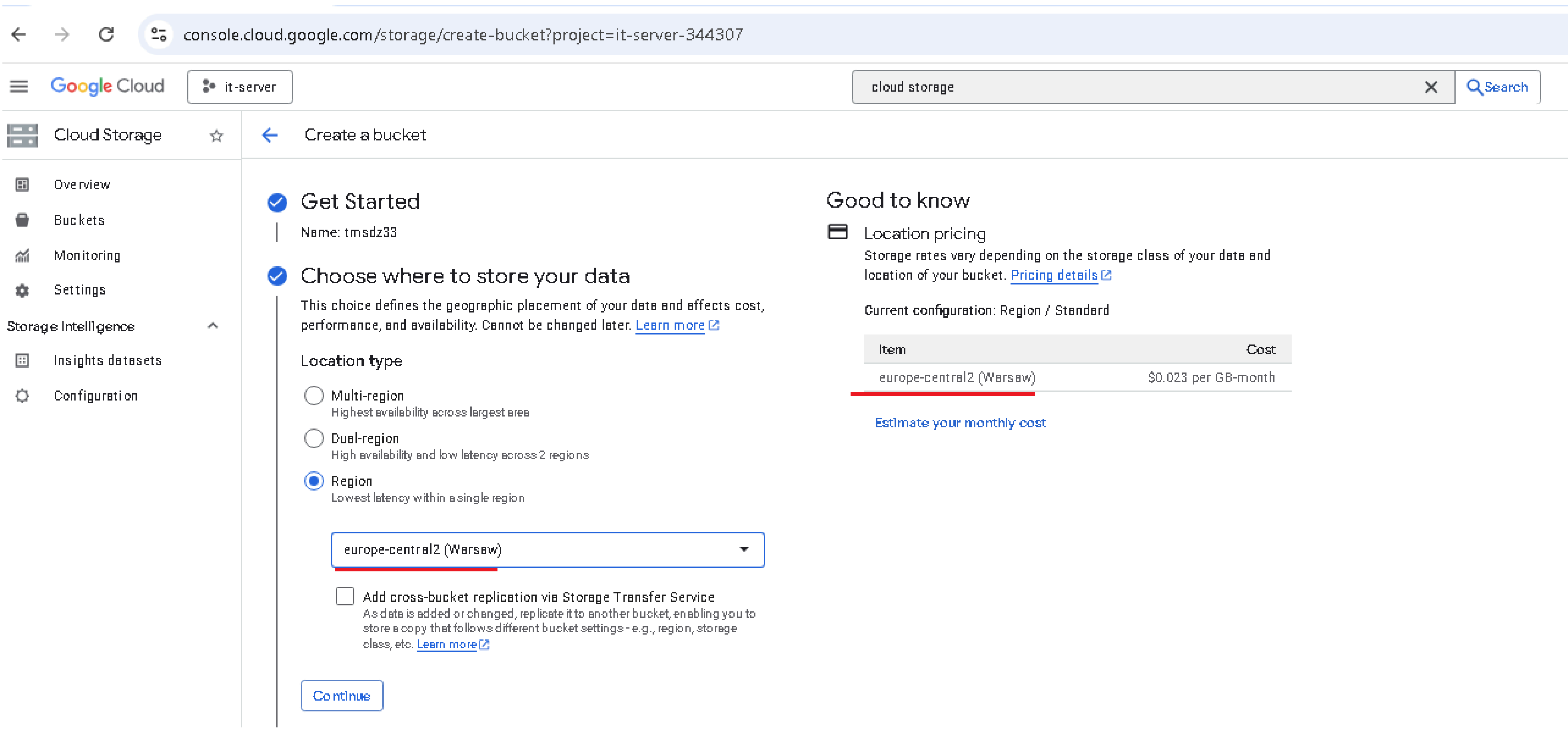
cloudshell:~ (it-server-344307)$ **gcloud storage buckets create gs://tmsdz33 --location=EUROPE-WEST2 --default-storage-class=STANDARD**

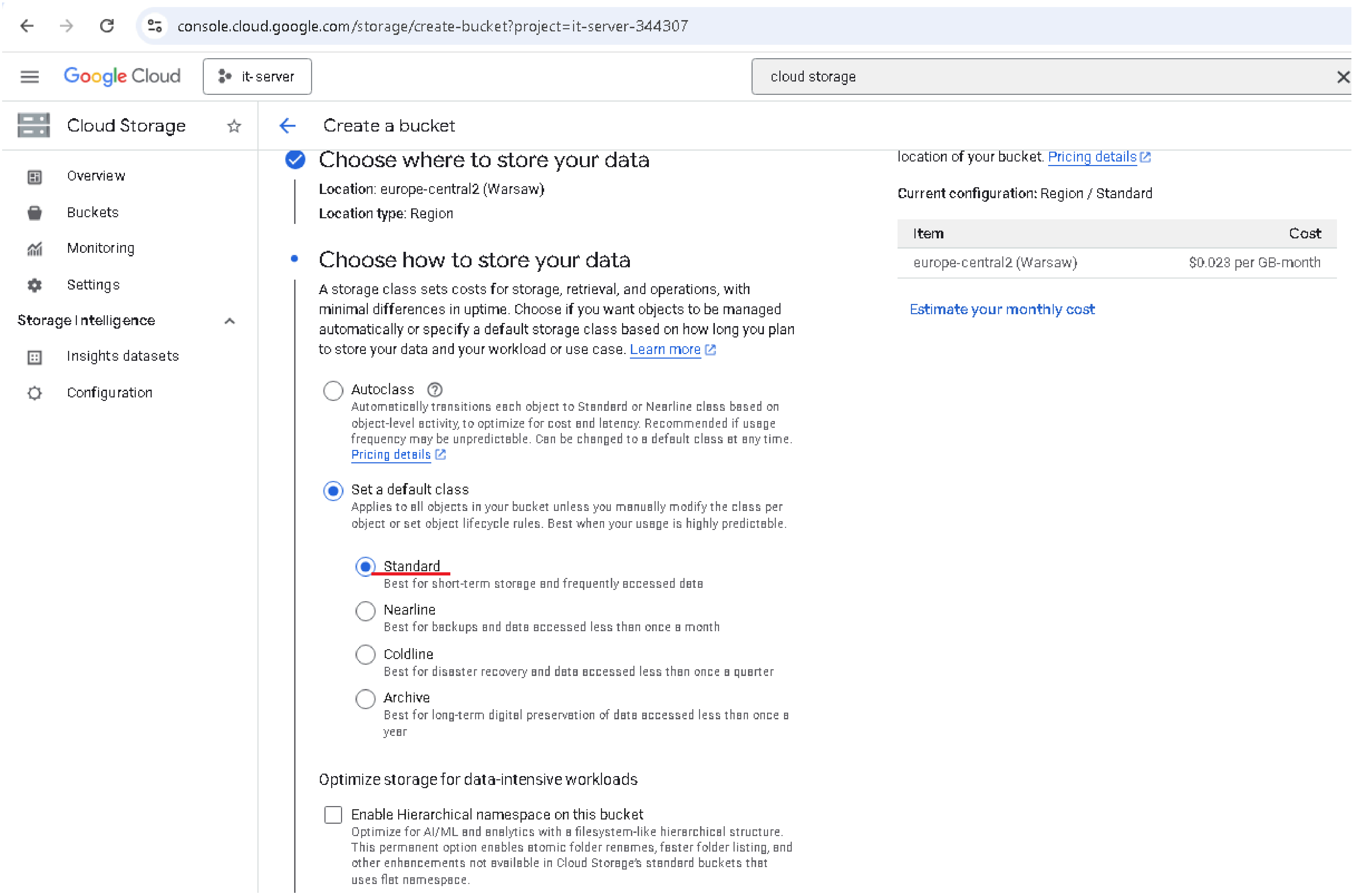
```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets create gs://tmsdz33      --location=EUROPE-WEST2      --default-storage-class=STANDARD
Creating gs://tmsdz33/...
```

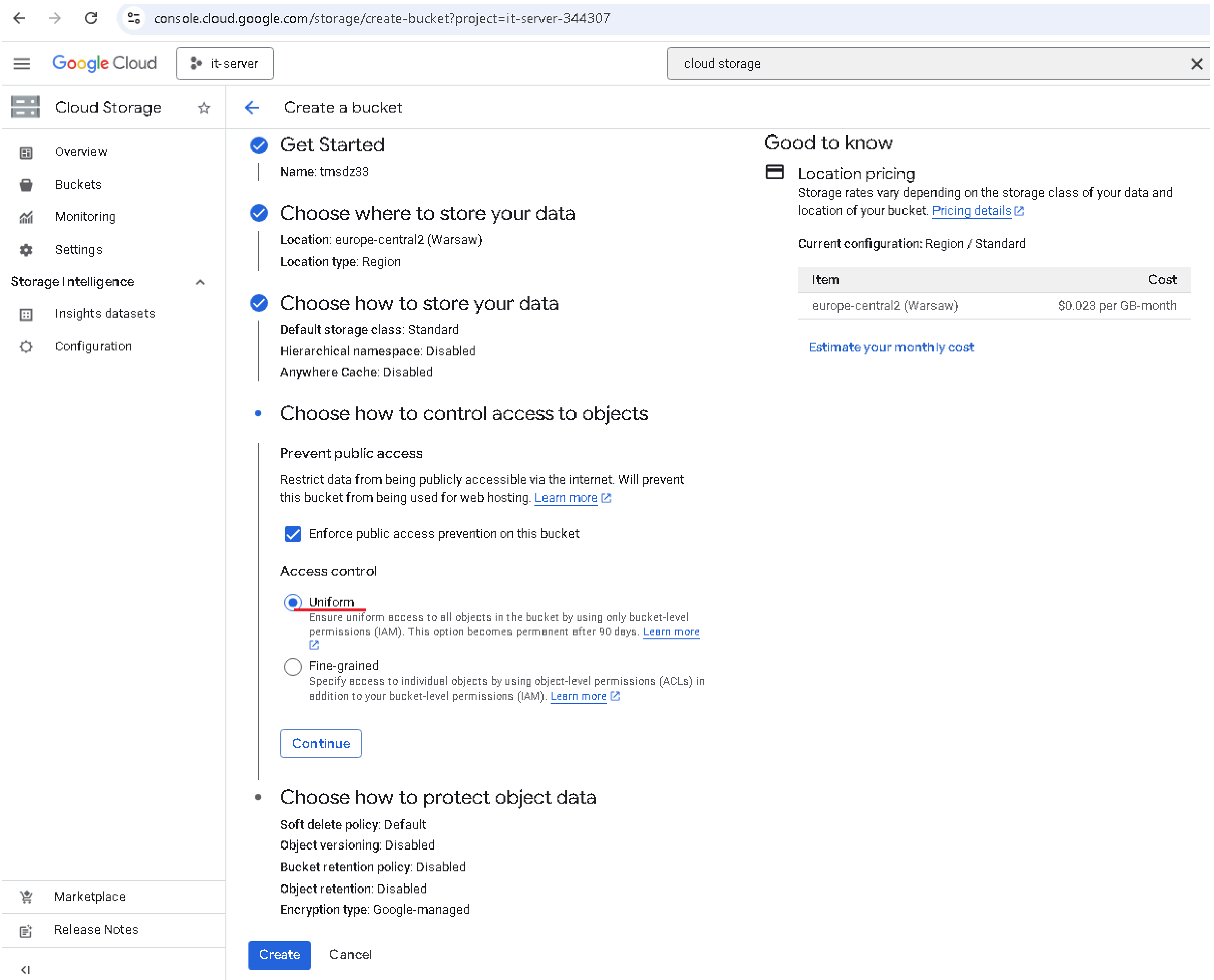
2.1) Создаем bucket (gui-вариант создания) :

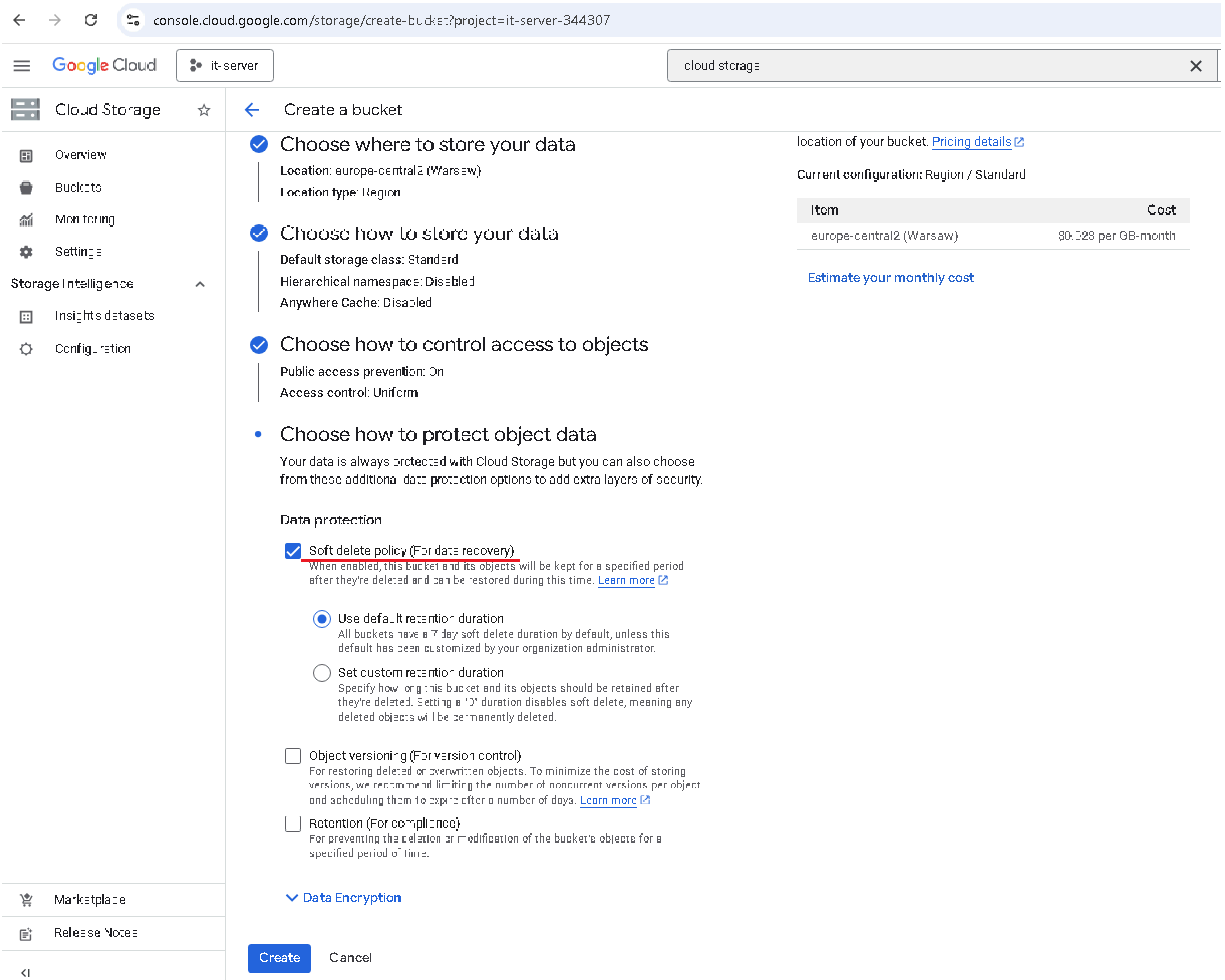



console.cloud.google.com/storage/create-bucket?project=it-server-344307
Google Cloud
it-server
cloud storage
Search
Cloud Storage
Create a bucket
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Get Started
Pick a globally unique, permanent name. Naming guidelines
tmsdz33
Tip: Don't include any sensitive information
Labels (optional)
Labels are key:value pairs that allow you to group related buckets together or with other Cloud Platform resources. Learn more
Key 1
descriptonbucket
Value 1
tmsdz33
Add label
Continue
Choose where to store your data
Location: us (multiple regions in United States)
Location type: Multi-region
Good to know
Location pricing
Storage rates vary depending on the storage class of your data and location of your bucket. Pricing details
Current configuration: Multi-region / Standard
Item
Cost
us (multiple regions in United States)
$0.026 per GB-month
With default replication
$0.020 per GB written
Estimate your monthly cost


console.cloud.google.com/storage/create-bucket?project=it-server-344307
Google Cloud
it-server
cloud storage
Search
Cloud Storage
Create a bucket
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Get Started
Name: tmsdz33
Choose where to store your data
This choice defines the geographic placement of your data and affects cost, performance, and availability. Cannot be changed later. Learn more
Location type
Multi-region
Highest availability across largest area
Dual-region
High availability and low latency across 2 regions
Region
Lowest latency within a single region
europe-central2 (Warsaw)
Add cross-bucket replication via Storage Transfer Service
As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc. Learn more
Continue
Good to know
Location pricing
Storage rates vary depending on the storage class of your data and location of your bucket. Pricing details
Current configuration: Region / Standard
Item
Cost
europe-central2 (Warsaw)
$0.023 per GB-month
Estimate your monthly cost


console.cloud.google.com/storage/create-bucket?project=it-server-344307
Google Cloud
it-server
cloud storage
Cloud Storage
Create a bucket
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Choose where to store your data
Location: europe-central2 (Warsaw)
Location type: Region
Choose how to store your data
A storage class sets costs for storage, retrieval, and operations, with minimal differences in uptime. Choose if you want objects to be managed automatically or specify a default storage class based on how long you plan to store your data and your workload or use case. Learn more
Autoclass
Automatically transitions each object to Standard or Nearline class based on object-level activity, to optimize for cost and latency. Recommended if usage frequency may be unpredictable. Can be changed to a default class at any time. Pricing details
Set a default class
Applies to all objects in your bucket unless you manually modify the class per object or set object lifecycle rules. Best when your usage is highly predictable.
Standard
Best for short-term storage and frequently accessed data
Nearline
Best for backups and data accessed less than once a month
Coldline
Best for disaster recovery and data accessed less than once a quarter
Archive
Best for long-term digital preservation of data accessed less than once a year
Optimize storage for data-intensive workloads
Enable Hierarchical namespace on this bucket
Optimize for AI/ML and analytics with a filesystem-like hierarchical structure. This permanent option enables atomic folder renames, faster folder listing, and other enhancements not available in Cloud Storage's standard buckets that uses flat namespace.
location of your bucket. Pricing details
Current configuration: Region / Standard
Item
Cost
europe-central2 (Warsaw)
$0.023 per GB-month
Estimate your monthly cost


console.cloud.google.com/storage/create-bucket?project=it-server-344307
Google Cloud
it-server
cloud storage
Cloud Storage
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Marketplace
Release Notes
Create a bucket
Get Started
Name: tmsdz33
Choose where to store your data
Location: europe-central2 (Warsaw)
Location type: Region
Choose how to store your data
Default storage class: Standard
Hierarchical namespace: Disabled
Anywhere Cache: Disabled
Choose how to control access to objects
Prevent public access
Restrict data from being publicly accessible via the internet. Will prevent this bucket from being used for web hosting. Learn more
Enforce public access prevention on this bucket
Access control
Uniform
Ensure uniform access to all objects in the bucket by using only bucket-level permissions (IAM). This option becomes permanent after 90 days. Learn more
Fine-grained
Specify access to individual objects by using object-level permissions (ACLs) in addition to your bucket-level permissions (IAM). Learn more
Continue
Choose how to protect object data
Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed
Create
Cancel
Good to know
Location pricing
Storage rates vary depending on the storage class of your data and location of your bucket. Pricing details
Current configuration: Region / Standard
Item
Cost
europe-central2 (Warsaw)
$0.023 per GB-month
Estimate your monthly cost



console.cloud.google.com/storage/create-bucket?project=it-server-344307
Google Cloud
it-server
cloud storage
Cloud Storage
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Marketplace
Release Notes
Create a bucket
Choose where to store your data
Location: europe-central2 (Warsaw)
Location type: Region
Choose how to store your data
Default storage class: Standard
Hierarchical namespace: Disabled
Anywhere Cache: Disabled
Choose how to control access to objects
Public access prevention: On
Access control: Uniform
Choose how to protect object data
Your data is always protected with Cloud Storage but you can also choose from these additional data protection options to add extra layers of security.
Data protection
Soft delete policy (For data recovery)
When enabled, this bucket and its objects will be kept for a specified period after they're deleted and can be restored during this time. Learn more
Use default retention duration
All buckets have a 7 day soft delete duration by default, unless this default has been customized by your organization administrator.
Set custom retention duration
Specify how long this bucket and its objects should be retained after they're deleted. Setting a '0' duration disables soft delete, meaning any deleted objects will be permanently deleted.
Object versioning (For version control)
For restoring deleted or overwritten objects. To minimize the cost of storing versions, we recommend limiting the number of noncurrent versions per object and scheduling them to expire after a number of days. Learn more
Retention (For compliance)
For preventing the deletion or modification of the bucket's objects for a specified period of time.
Data Encryption
Create
Cancel
location of your bucket. Pricing details
Current configuration: Region / Standard
Item
Cost
europe-central2 (Warsaw)
$0.023 per GB-month
Estimate your monthly cost

2.2) Предоставляем доступ к бакету " tmsdz33" по email :

cloudshell:~ (it-server-344307)$ **gcloud storage buckets add-iam-policy-binding gs://tmsdz33   --member="user:egor.belousov@altezza.org" --role="roles/storage.objectAdmin"**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets add-iam-policy-binding gs://tmsdz33   --member="user:egor.belousov@altezza.org"   --role="roles/storage.objectAdmin"
bindings:
- members:
  - projectEditor:it-server-344307
  - projectOwner:it-server-344307
  role: roles/storage.legacyBucketOwner
- members:
  - projectViewer:it-server-344307
  role: roles/storage.legacyBucketReader
- members:
  - user:egor.belousov@altezza.org
  role: roles/storage.objectAdmin
etag: CAU=
kind: storage#policy
resourceId: projects/_/buckets/tmsdz33
version: 1
egor_belousov@cloudshell:~ (it-server-344307)$
```

2.3) Проверяем атрибуты нашего бакета "tmsdz33" :

cloudshell:~ (it-server-344307)$ **gcloud storage buckets describe gs://tmsdz3**

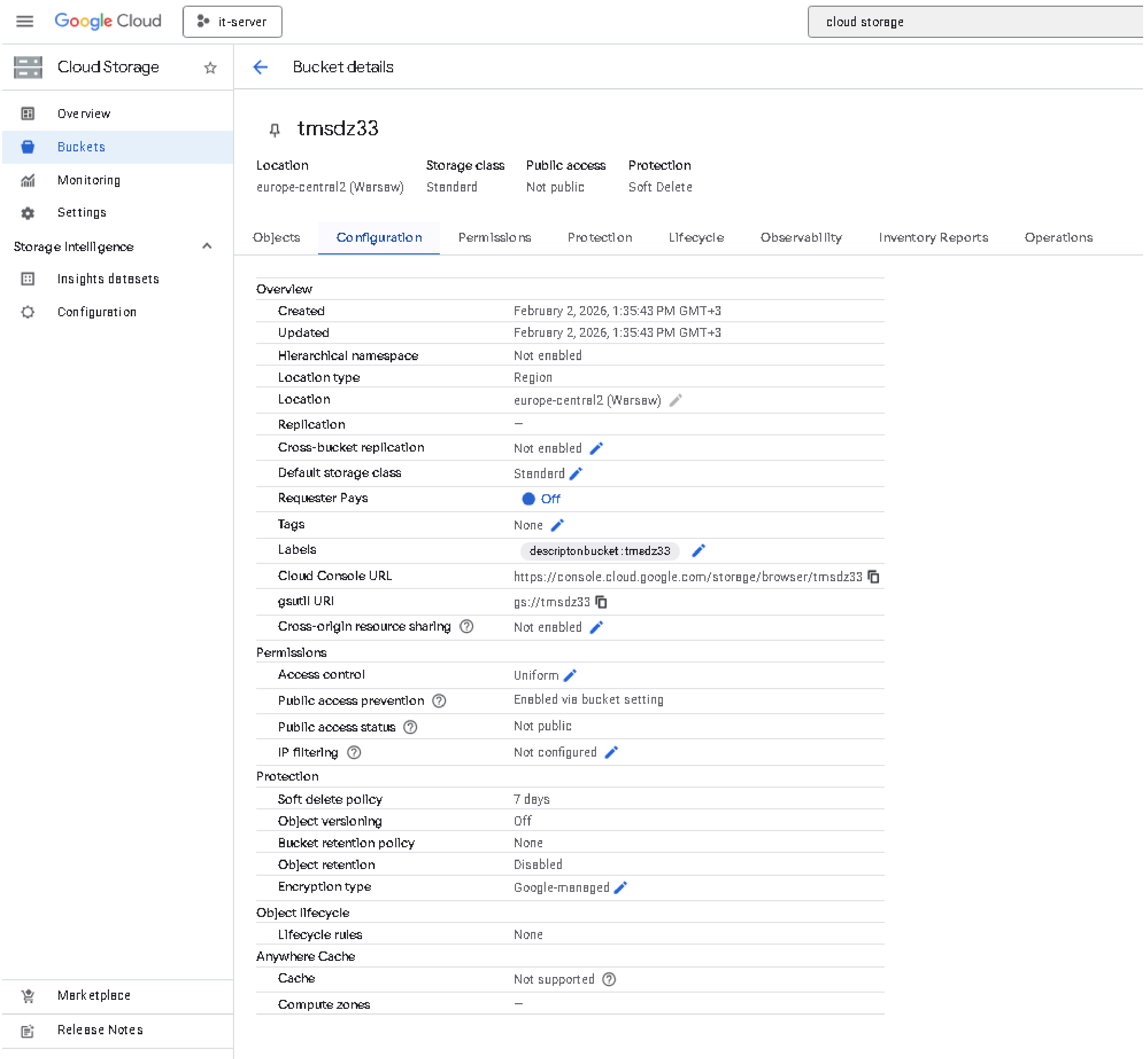
```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz33
acl:
- entity: project-viewers-52615052819
  projectTeam:
    projectNumber: '52615052819'
    team: viewers
  role: READER
default_storage_class: STANDARD
generation: 1770038782129218882
location: EUROPE-WEST2
location_type: region
metageneration: 5
name: tmsdz33
public_access_prevention: inherited
satisfies_pzs: true
soft_delete_policy:
  effectiveTime: '2026-02-02T13:26:22.487000+00:00
  retentionDurationSeconds: '604800'
storage_url: gs://tmsdz33/
uniform_bucket_level_access: false
update_time: 2026-02-02T17:53:24+0000
egor_belousov@cloudshell:~ (it-server-344307)$
```

// можно смотреть атрибуты бакета “ tmsdz33 “ в json-формате:

cloudshell:~ (it-server-344307)$ **gcloud storage buckets describe gs://tmsdz33 --format="json(name,location,storageClass)"**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz33 --format="json(name,location,storageClass)"
{
  "location": "EUROPE-WEST2",
  "name": "tmsdz33"
}
egor_belousov@cloudshell:~ (it-server-344307)$
```

2.4) Проверяем свойства бакета через GUI



3) Подключаемся к google cloud s3 bucket с ubuntu22 :

ubuntu22# **snap install google-cloud-cli --classic**   // **пакет для работы с Google Cloud API на ubuntu**
ubuntu22# **gcloud auth login**   // **выдет https-строку-запрос-на-токен идем в браузер и авторизация через свою email-учетку.**

```
-white:~# gcloud auth login
Go to the following link in your browser, and complete the sign-in prompts:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=htt
ml&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform
engine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlser                                              compute+https%3
auth&state=KGLvJN9b2EPGi03iq3xxssLewcCDM5&prompt=consent&toke                                              DL1vfztIBUwtGif
d=S256

Once finished, enter the verification code provided in your browser: 4/0ASc3gC3DsbwMgPhwlooqekMJ3sT4gMqAfrO8WdGGe523Lrx5tfdH-IWqn

You are now logged in as [                    ].
Your current project is [it-server-344307].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
```

P.S.#1 Можно и без браузера через "**gcloud auth login --no-browser** " но до конца не разобрался как.

ubuntu22# **gcloud config set project it-server-344307**   // где it-server-344307 - ID проекта).

```
-white:~# gcloud config set project it-server-344307
Project 'it-server-344307' lacks an 'environment' tag. Please create or add a tag with key 'environment' and a value like 'Production', 'Development', 'Test', or 'Staging'. Ad
d an 'environment' tag using `gcloud resource-manager tags bindings create`. See https://cloud.google.com/resource-manager/docs/creating-managing-projects#designate_project_en
vironments_with_tags for details.
Updated property [core/project].
```

ubuntu22#**gcloud storage buckets create gs://tmsdz331 --location="europe-west2"**
Creating gs://tmsdz331/...

3.1) Проверяем какие бакеты создались через cli :

Buckets   Create   Refresh

Filter   Filter buckets

Name ↑	Created	Location type	Location	Default storage class	Last modified	Public access
tmsdz33	Feb 2, 2026, 4:26:22 PM	Region	europe-west2	Standard	Feb 3, 2026, 12:13:15 AM	Subject to object ACLs
tmsdz331	Feb 8, 2026, 11:40:22 PM	Region	europe-west2	Standard	Feb 8, 2026, 11:40:22 PM	Subject to object ACLs

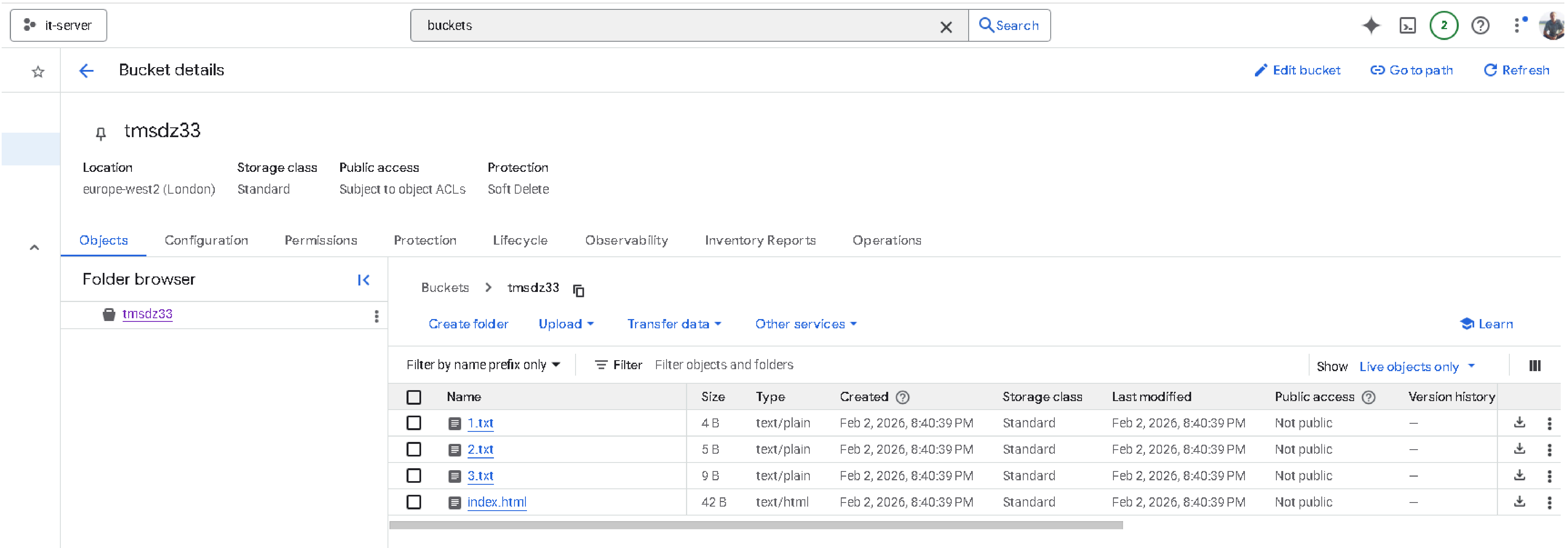
3.2) Копируем все файлы из папки /home/gitlab-runner/testsite/   в   бакет “ tmsdz33 “   :

ubuntu22# **gcloud storage cp * gs://tmsdz33**

```
root@U22GITLABNODE2:/home/gitlab-runner/testsite#
root@U22GITLABNODE2:/home/gitlab-runner/testsite# nano 1.txt
root@U22GITLABNODE2:/home/gitlab-runner/testsite# nano 2.txt
root@U22GITLABNODE2:/home/gitlab-runner/testsite# nano 3.txt
root@U22GITLABNODE2:/home/gitlab-runner/testsite# gcloud storage cp * gs://tmsdz33
Copying file://1.txt to gs://tmsdz33/1.txt
Copying file://2.txt to gs://tmsdz33/2.txt
Copying file://3.txt to gs://tmsdz33/3.txt
Copying file://index.html to gs://tmsdz33/index.html
  Completed files 4/4 | 60.0B/60.0B

Average throughput: 247.6B/s
root@U22GITLABNODE2:/home/gitlab-runner/testsite#
```

3.3)Проверяем в GUI GCP что файлы успешно скопировались в бакет:



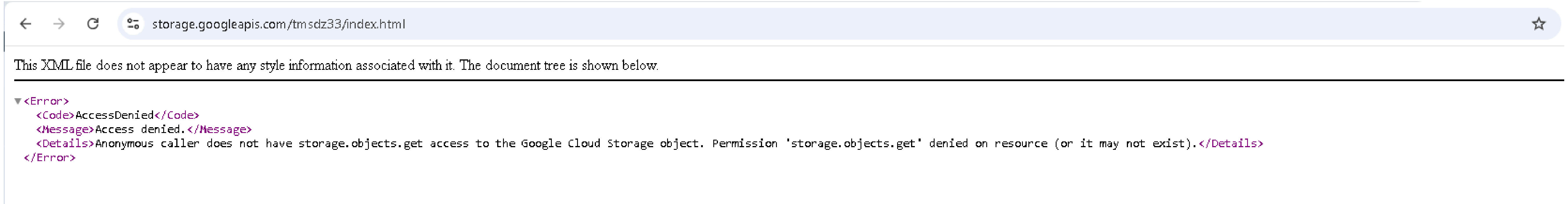
3.4) **Копируем файлы во 2й бакет "tmsdz331" :**

ubuntu22# gcloud storage cp * gs://tmsdz331

```
root@U22GITLABNODE2:/home/gitlab-runner/testsite# gcloud storage cp * gs://tmsdz331
Copying file://1.txt to gs://tmsdz331/1.txt
Copying file://2.txt to gs://tmsdz331/2.txt
Copying file://3.txt to gs://tmsdz331/3.txt
Copying file://index.html to gs://tmsdz331/index.html
  Completed files 4/4 | 60.0B/60.0B

Average throughput: 214.6B/s
```

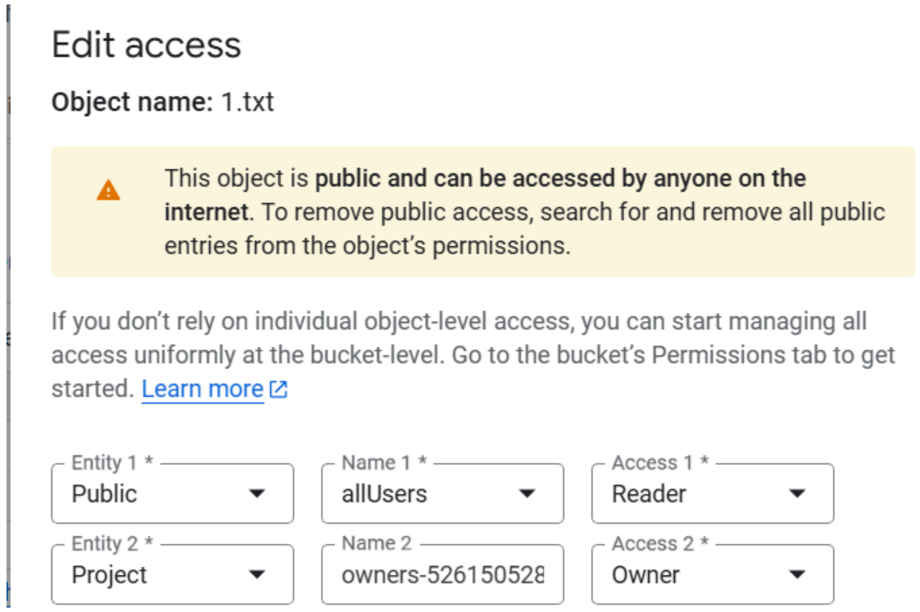
3.5) Проверяем доступ из Интернета, его нет , так как при создании указали бакет без доступа из Интернета (без public доступа) поэтому все норм, доступ только через key.json.

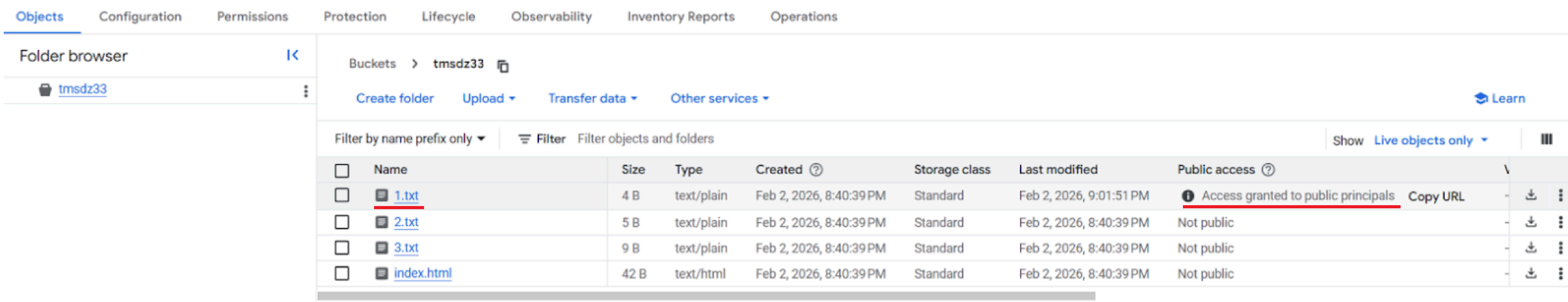


3.6) Открываем доступ из Интернета к конкретному файлу в S3 bucket :

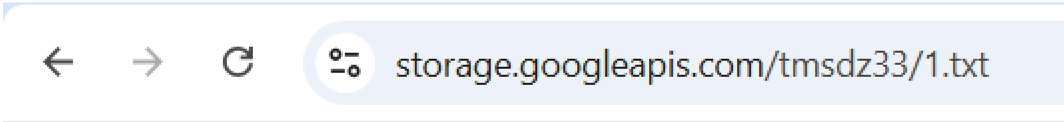
cloudshell:~ (it-server-344307)$ $ gsutil acl ch -u AllUsers:R gs://tmsdz33/1.txt

//Через gui GCP:





3.7) **Проверяем в браузере доступ к файлу 1.txt (открыт на чтение для всего Интернета) :**



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3.8) Удаляем из бакета s3://tmsdz33 файл 3.txt

```
root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com   --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]

root@U22GITLABNODE2:/home/gitlab-runner# gcloud storage rm gs://tmsdz33/3.txt
Removing objects:
Removing gs://tmsdz33/3.txt...
  Completed 1/1
```

```
root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com   --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
root@U22GITLABNODE2:/home/gitlab-runner#
root@U22GITLABNODE2:/home/gitlab-runner#
root@U22GITLABNODE2:/home/gitlab-runner#
root@U22GITLABNODE2:/home/gitlab-runner# gcloud storage rm gs://tmsdz33/3.txt
Removing objects:
Removing gs://tmsdz33/3.txt...
  Completed 1/1
root@U22GITLABNODE2:/home/gitlab-runner#
```

Используемые команды :

1) Создание сервисного аккаунта с ролью Storage.Admin:

cloudshell:~ (it-server-344307)$ gcloud iam service-accounts create sa-tmsdz33 --display-name "TMS service account"
cloudshell:~ (it-server-344307)$ gcloud projects add-iam-policy-binding $DEVSHELL_PROJECT_ID --member serviceAccount:sa-tmsdz33@$DEVSHELL_PROJECT_ID.iam.gserviceaccount.com --role="roles/storage.admin"
cloudshell:~ (it-server-344307)$ PROJECT_ID="$DEVSHELL_PROJECT_ID"
cloudshell:~ (it-server-344307)$ SA_NAME="sa-tmsdz33"
cloudshell:~ (it-server-344307)$ gcloud config set project it-server-344307
// указываем GCP- проект
cloudshell:~ (it-server-344307)$ gcloud iam service-accounts keys create ~/keys/${SA_NAME}.json --iam-account=${SA_NAME}@${PROJECT_ID}.iam.gserviceaccount.com
P.S.#1 или (без переменных) : gcloud iam service-accounts keys create key.json --iam-account=gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com

1.3) Проверяем созданные IAM-учетки в нашем проекте : cloudshell:~ (it-server-344307)$ gcloud iam service-accounts list
1.4) копируем ключ (json-файл) на удаленный хост : cloudshell:~/keys (it-server-344307)$ cat /home/user/keys/sa-tmsdz33.json
1.5) авторизоваться на ubuntu22 по key.json-файлу : ubuntu22# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com --key-file=key.json
2) Создаем S3 GCP Bucket (cli-вариант) : cloudshell:~ (it-server-344307)$ gcloud storage buckets create gs://tmsdz33 --location=EUROPE-WEST2 --default-storage-class=STANDARD
2.2) Предоставляем доступ к бакету " tmsdz33" по email : cloudshell:~ (it-server-344307)$ gcloud storage buckets add-iam-policy-binding gs://tmsdz33 --member="user:egor.belousov@altezza.org" --role="roles/storage.objectAdmin"
2.3) Проверяем атрибуты нашего бакета "tmsdz33" : cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz3
P.S.#2 Проверяем атрибуты нашего бакета "tmsdz33" в json-формате : cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz33 --format="json(name,location,storageClass)"

3) Подключаемся к google cloud s3 bucket с ubuntu22 :
ubuntu22# snap install google-cloud-cli --classic
ubuntu22# gcloud auth login // выдет https-строку-запрос-на-токен идем в браузер и авторизация через свою email-учетку.
P.S.#3 Можно и без браузера через "gcloud auth login --no-browser " но до конца не разобрался как.
ubuntu22# gcloud config set project it-server-344307 // где it-server-344307 - ID проекта)
ubuntu22# gcloud storage buckets create gs://tmsdz331 --location="europe-west2"
ubuntu22# gcloud storage cp * gs://tmsdz33
ubuntu22# gcloud storage cp * gs://tmsdz331 //Копируем файлы во 2й бакет "tmsdz331"

3.6) Открываем доступ из Интернета к конкретному файлу в S3 bucket :
cloudshell:~ (it-server-344307)$ $ gsutil acl ch -u AllUsers:R gs://tmsdz33/1.txt

3.8) Удаляем из бакета "tmsdz33 " файл 3.txt :
ubuntu22 # gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com --key-file=key.json
ubuntu22 # gcloud storage rm gs://tmsdz33/3.txt